

Why Sports Betting Isn't Profitable

Even With AI (I Tested It)

By Stefan Soh

The Question I Wanted to Answer

Can everyday people profit from sports betting?

THE SETUP:

- Sports betting is now legal in most states
- Apps make it look simple: 'Just pick the winner!'
- But studies show 95% of bettors lose money

MY HYPOTHESIS:

If I can't beat it with AI and data, regular people definitely can't.

How I Tested It

THE DATA:

- 1,226 NBA games (2018-2019 season)
- 2 million individual shot records
- Vegas betting lines (spreads, totals, moneylines)
- Team statistics and performance metrics

THE APPROACH:

- Built AI models to predict if teams would 'cover the spread'
- Simulated 1,000 bets to test profitability

Spread betting: betting if a team wins by MORE than the predicted margin

- Analyzed where Vegas pricing comes from

What is Spread Betting?

EXAMPLE:

Lakers vs Celtics

Vegas spread: Lakers -7.5

This means:

- Lakers are favored to win by 7.5 points
- If you bet Lakers -7.5, they must win by 8+ for you to win
- If Lakers win by 7 or less (or lose), you lose the bet

WHY IT'S HARD:

Test #1: Can AI Predict Spread Outcomes?

72.36%

Accuracy on Test Games

- ✓ Trained on 980 games from 2018-19 season
- ✓ Tested on 246 games the model had never seen
- ✓ Correctly predicted 178 out of 246 test games
- ✓ Better than random guessing (50%)
- ✓ Better than typical sports bettors (~52%)

The Hidden Cost: The 'Vig'

To win \$100, you must risk \$110

$$\text{\$110} \div \text{\$210} = 52.38\%$$

You need to win 52.38% of bets just to break even
(NOT 50% like most people think)

This 2.38% gap is the house edge that kills you over time

Test #2: Simulating 1,000 Spread Bets

I simulated betting \$110 on 1,000 spread bets at 72% accuracy:

WHAT YOU'D EXPECT:

- Win 720 bets $\times$ \$100 profit = \$72,000
- Lose 280 bets $\times$ \$110 risk = -\$30,800
- Net profit: +\$41,200 ✓

WHAT ACTUALLY HAPPENS:

- Win 720 bets $\times$ \$100 = \$72,000

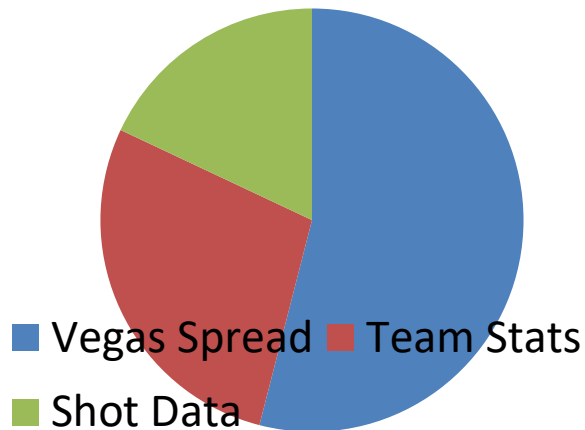
This assumes you can bet ALL 1,000 games at 72% accuracy

- Lose 280 bets $\times$ \$110 = -\$30,800

Test #3: How Does Vegas Price Games?

DISCOVERY:
Vegas is INCREDIBLY accurate

Model Predictive Power



The Vegas spread was 54% of what made my model accurate.
My 'AI edge' was only the remaining 46% from shot data and team stats.

How Risk Affects Betting Odds

Vegas adjusts payouts based on how risky the bet is:

EASY BET (Heavy Favorite):

- Lakers -12.5 vs terrible team
- Very likely to cover (high probability)
- Payout: Risk \$500 to win \$100
- Less risk = less reward

HARD BET (Underdog):

- Underdog +12.5 vs Lakers

Test #4: Can I Find "Good" Bets?

I simulated 1,000 spread bets looking for positive expected value opportunities:

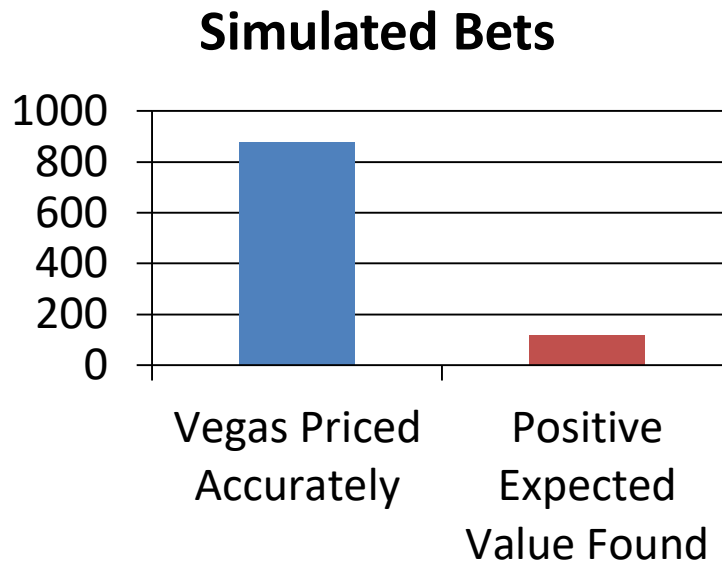
RESULT:

Only 120 bets (12%) had positive expected value

These are games where Vegas slightly mispriced the spread.

Betting ONLY these 120:

- Win rate: 58.3%
- Profit: +\$5,410



Why Even "Good" Bets Don't Work in Practice

THE FOUR BARRIERS:

1. SPORTSBOOKS BAN WINNERS

- If you consistently win, they limit your bet size or ban you
- This happens within 3-6 months for documented winners

2. OPPORTUNITIES ARE RARE

- Only 12% of bets = ~150 games per NBA season
- Need \$15,000+ in capital sitting idle for 6 months
- Profit of \$5,410 on \$15K capital = 36% annual return (sounds good...)

How The House Stays Profitable

THREE ADVANTAGES VEGAS HAS:

1. ACCURATE PRICING

- Professional oddsmakers with decades of data
- Aggregate information from thousands of bets
- Real-time adjustments as money flows in

2. THE VIG IS GUARANTEED PROFIT

- Even when Vegas is slightly wrong on a spread, the 4.5% vig covers it
- They don't need perfect predictions, just close enough

What I Learned

- ✗ Even 72% AI accuracy isn't enough
- ✗ Finding 'good' bets (12%) isn't enough
- ✗ Being smarter than other bettors isn't enough
- ✓ The vig (4.5%) erodes profits systematically
- ✓ Vegas prices spreads with 54% of my model's accuracy
- ✓ Variance kills individuals, volume protects the house
- ✓ Winners get banned before making serious money

BOTTOM LINE:

Sports betting isn't about being smart or having better data.
It's a mathematically losing game designed to profit the house.